Erik Löffelholz

Leipzig, Germany eriklfholz@googlemail.com github.com/erik2810 erik2810.github.io

PERSONAL PROFILE & RESEARCH INTERESTS

Master's graduate in Mathematical Physics with a strong foundation in continuous–discrete interactions and differential geometry. Currently specializing in Graph Machine Learning and physics-inspired generative modeling. Passionate about understanding how complex structure emerges from local constraints by integrating variational principles, energy-based modeling, and structural priors. Applied experience spans joint discrete-continuous diffusion models, differentiable physics simulation, rigorous metric evaluation, and from-scratch gradient-based optimization.

EDUCATION

Oct 2021 – Sep 2025

M.Sc. Mathematical Physics
Universität Leipzig — Faculty of Physics and Earth System Sciences
Selected Coursework: Advanced PDE and Analysis · Differential Geometry · Quantum Field Theory · General Relativity · Group Theory

Oct 2018 – Mar 2022

B.Sc. Physics
Universität Leipzig — Faculty of Physics and Earth System Sciences

Aug 2010 – Jun 2018

Abitur
Luther-Melanchthon-Gymnasium, Lutherstadt Wittenberg

RESEARCH EXPERIENCE

Apr 2025 – Present

Scientific Assistant — Mathematics Lab
Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany

- Developed differentiable simulation software and 3D visualization tools
- Implemented mathematical models in Python and PyTorch
- Transitioned from theoretical physics modeling to computational research

2025 – Present

Independent Researcher & Developer
Graph ML, Computational Geometry & Differentiable Physics

TEACHING & ACADEMIC SERVICE

Apr 2025 – Present

Mathematics Lab Assistant (Wissenschaftliche Hilfskraft)
Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany

- Supported the Mathematics Lab, facilitating practical mathematical applications and assisting in the execution of technical modeling tasks.

Mar 2024 – Feb 2025

Working Student — Mathematics Editorial
Ernst Klett Verlag GmbH, Leipzig

- Conducted technical and editorial reviews of mathematical content, ensuring structural clarity and pedagogical accuracy for educational publications.

PUBLICATIONS & CONFERENCES

Fabian Lander, **Erik Löffelholz**, Diaaeldin Taha, Steve Trettel, Anna Wienhard.
"Illustrating Hyperbolic Surfaces with Mesh Embeddings."
Submitted to the **Bridges Conference 2026** (Regular Papers Track). *Under Review.*

RELEVANT SKILLS & LANGUAGES

Programming
Python · JavaScript · TypeScript · C++

Machine Learning
PyTorch (Autograd, PINNs, GNNs) · Diffusion Models · VAEs · Neural ODEs

Scientific Computing
Numerical Integration (Runge-Kutta, Monte Carlo) · NumPy · SciPy

Mathematical Methods
Functional analysis · PDE theory · variational methods · differential geometry · group theory · Monte Carlo integration

Tools
Git · Docker · FastAPI · LaTeX

Languages
German (native) · English (C1 — full professional proficiency)

REFEREES

Dr. Diaaeldin Taha
Research Group Leader (Mathematical Structures in AI)
Max Planck Institute for Mathematics in the Sciences, Leipzig
taha@mis.mpg.de

SELECTED PROJECTS

Graph ML Lab — From-scratch implementations of GCN, GAT, graph VAE, discrete diffusion, and joint discrete-continuous diffusion over graph structure and 3D positions. PyTorch only, no external GNN libraries. github.com/erik2810/ml-projects

DiffQFT — Differentiable quantum field theory computations in Euclidean AdS_2. Witten diagram integration, neural surrogates, PINN solver for the Klein-Gordon equation. github.com/erik2810/DiffQFT

Differentiable Physics Engine — Neural ODE learning chaotic Lorenz dynamics from scratch in the browser. From-scratch backpropagation and Adam optimizer in JavaScript. github.com/erik2810/differentiable-physics-engine

Full project portfolio with live demos at erik2810.github.io/projects